

Supplementary Material for ForeTac: Steering Robot Actions with Predicted Contact Consequences

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TABLE I
HDF5 EPISODE FIELDS USED BY THE CURRENT IMPLEMENTATION.

Field	Shape / type	Usage
observations/images/global	$T \times H \times W \times 3$ uint8	scene context
observations/images/wrist	$T \times H \times W \times 3$ uint8	local contact geometry
observations/proprio_joint	$T \times 7$ float	current robot state
actions/joint_abs	$T \times 7$ float	action chunk target / foresight condition
observations/tac/left/marker_offset	$T \times 9 \times 9 \times 2$ float	GelSight marker displacement
observations/tac/left/force6d	$T \times 6$ float	contact-quality labels and analysis

I. APPENDIX OVERVIEW

This appendix provides implementation details and additional evidence for ForeTac. Section II describes the real-robot data format and the current local data inventory. Section III gives training and deployment hyperparameters. Section IV expands the algorithmic pipeline. Sections V–?? report the tactile encoder, foresight, scorer, force-curve, main-comparison, and ablation results. Some controlled on-line rollouts and qualitative image panels are still being collected; entries that are not yet verified are left as TBD.

II. DATA COLLECTION AND FORMAT

A. Robot Observation Schema

All policies and foresight modules are trained from timestamp-aligned HDF5 episodes. A trajectory contains synchronized RGB views, proprioception, action labels, tactile marker fields, and optional force/torque measurements. The main action-conditioned foresight contract is

$$\hat{\mathbf{z}}_{t:t+H-1} = f_\psi(I_t^g, I_t^w, \mathbf{m}_{t-W+1:t}, q_t, \mathbf{a}_{t:t+H-1}), \quad (1)$$

where the condition is the future action chunk `actions/joint_abs[t:t+H]`, not the future state trajectory. This distinction is important: future states are outcomes, while the rollout-time quantity available before execution is the candidate action chunk.

RGB images are resized and cropped before ImageNet normalization. The tactile input is a GelSight marker displacement field, normalized per displacement channel. TacVAE consumes an eight-frame tactile window and encodes the current contact into a compact spatial latent. The diffusion policy uses `obs_horizon=2` and predicts a 16-step action chunk for the main experiments.

B. Current Local Data Inventory

Table II lists the data currently visible in the local workspace and used by the corresponding training/evaluation scripts. The board split is the labeled board TacVAE split; card and vase splits are the current task-specific TacVAE splits; insertion uses the truncated insertion datasets for policy and score-model development.

C. Task Labels and Safety Definitions

For each online trial, a success is counted when the task objective is completed within the rollout horizon. A safe success is stricter: the task must succeed without contact-force violations, object disturbance, emergency stops, or task-specific unsafe modes. The task-specific safety rules are:

- Board wiping: stable moderate force over the wipe path, with no long contact dropout and no force spike above the allowed band.
- Vase wiping: continuous contact on the curved surface, low object motion, and no tip-over or high-force spike.
- Card swiping: sustained slot contact and successful pass-through without dropout or large misalignment.
- Chip grasping: lift without crushing or slipping.
- Socket insertion: insertion success with bounded peak force, limited retry count, and no sustained jamming.

III. TRAINING DETAILS

A. Model Components

Table III summarizes the implementation settings used in the current experiments. These values match the local scripts and configs used for the project page and appendix evidence.

B. Foresight Loss

The foresight model predicts the latent sequence corresponding to the candidate action chunk. Training uses frozen TacVAE targets and combines three terms:

$$\mathcal{L}_{\text{fs}} = \mathcal{L}_{\text{latent}} + \lambda_m \mathcal{L}_{\text{marker}} + \lambda_\Delta \mathcal{L}_\Delta, \quad (2)$$

$$\mathcal{L}_{\text{latent}} = \text{SmoothL1}(\hat{\mathbf{z}}_{1:H}, \mathbf{z}_{1:H}) + \text{SmoothL1}(\hat{\mathbf{z}}_H, \mathbf{z}_H), \quad (3)$$

$$\mathcal{L}_{\text{marker}} = \text{SmoothL1}(D_{\text{tac}}(\hat{\mathbf{z}}_{1:H}), \mathbf{m}_{1:H}), \quad (4)$$

$$\mathcal{L}_\Delta = \text{SmoothL1}(\Delta \hat{\mathbf{z}}_{1:H-1}, \Delta \mathbf{z}_{1:H-1}). \quad (5)$$

The main configuration uses $\lambda_m = 0.3$ and $\lambda_\Delta = 0.5$. Decoded marker loss keeps the latent forecast physically tied to marker displacement, while the delta term penalizes temporally inconsistent predictions.

C. Correct Action Conditioning

The current corrected training path sets `use_state_trajectory=false`. For a sample starting at time t , the foresight input is `actions/joint_abs[t:t+H]` and the target is the tactile future `marker_offset[t+1:t+H]` or the stride-aligned equivalent. The previous state-trajectory mode is kept only as a legacy option for ablations and should not be used for the main ForeTac claim.

TABLE II
CURRENT LOCAL EPISODE INVENTORY. COUNTS ARE FROM THE LOCAL HDF5 DIRECTORIES AND SAVED SPLIT METRICS.

Task	Main local sources	Episodes / split	Primary role
Board wiping	260609/wipe_pos, z_too_high, 260610/z_too_low, z_too_oscillate	177 train / 44 test; 9631 test windows	TacVAE, force-band scorer, action-conditioned foresight
Board wiping extra positives	260617_v81_caheiban/peg_in_hole_0617	80 episodes	positive contact reference and rollout validation
Vase wiping	260630_v8j_huaping/peg_in_hole_0630	90 train / 10 test; 2088 test windows	TacVAE and contact-stability analysis
Card swiping	260615_v81_card, 260626_replay_card, 260629/260701_v8j_card	246 train / 27 test; 6033 test windows	TacVAE and card positive/negative data analysis
Chip grasping	TBD	TBD	fragile grasp rollout table placeholder
Socket insertion	0209-0210_truncated, 260401_k14_truncated	318 + 162 episodes visible locally	insertion policy/scorer development and 20-trial benchmark

TABLE III
MAIN TRAINING HYPERPARAMETERS.

Component	Architecture / input	Optimization	Notes
TacVAE	8-frame $9 \times 9 \times 2$ marker windows; latent $16 \times 3 \times 3$	AdamW, lr 10^{-4} , batch 512, 150 epochs, stride 2 or 4	KL weight 10^{-6} ; direction loss weight 0.2
Visual/tactile DP	global+wrst RGB, TacVAE latent history, q_t ; action horizon 16	AdamW, lr 10^{-4} , batch 64, 1000 epochs for board	DDPM training steps 100; inference steps 100; action space joint_abs
Action-conditioned foresight	ResNet18 visual tokens, TacVAE tokens, q_t ; action chunk $H = 16$	AdamW, lr 4×10^{-5} , weight decay 10^{-4}	3 transformer layers, 8 heads, hidden 512, FPN 2048
Contact-quality scorer	predicted latent sequence and optional action/force features	cross-entropy / margin objective	deployed score is expert-good margin against negative modes
Guided denoising	late DDPM steps only	normalized gradient step; trust-region clipping	action branch detached in latent-only path

IV. ALGORITHMIC PIPELINE

Algorithm S1: Inference-time predicted-contact guidance.

- 1) Read the current observation $\mathbf{o}_t = (I_t^g, I_t^w, q_t, \mathbf{m}_{t-W+1:t})$.
- 2) Encode the tactile history with TacVAE to obtain the current latent $\mathbf{z}_t$ and tactile tokens.
- 3) Run the base action generator to sample a denoising trajectory for an action chunk $\mathbf{a}_{t:t+H-1}$.
- 4) At selected late denoising steps, convert the noisy action sample to the clean action estimate $\hat{\mathbf{x}}_0$.
- 5) Feed $\hat{\mathbf{x}}_0$ into the action-conditioned foresight model and predict the future tactile latent sequence $\hat{\mathbf{z}}_{t:t+H_f-1}$.
- 6) Score the predicted tactile future with the contact-quality energy model:

$$s_{\text{margin}} = \ell_{\text{good}} - \log \sum_{c \in \mathcal{C}_{\text{bad}}} \exp(\ell_c). \quad (6)$$

- 7) Backpropagate s_{margin} through $\mathbf{a} \rightarrow \hat{\mathbf{z}} \rightarrow S$ and apply a normalized small update inside a raw-action trust region.
- 8) Execute only the first action steps of the refined chunk and repeat at the next control cycle.

This design separates behavior generation from contact-quality correction. The base policy remains the prior over demonstrated robot motions; foresight asks what tactile consequence a candidate chunk would create; the scorer decides whether that predicted future contact is desirable.

V. TACTILE ENCODER AND RECONSTRUCTION

A. Encoder Construction

TacVAE uses an encoder-decoder structure over marker displacement fields rather than raw tactile images. The input is an eight-frame marker window; the encoder maps it to a $16 \times 3 \times 3$ spatial latent and the decoder reconstructs a $9 \times 9 \times 2$ marker field. The spatial latent keeps local contact geometry available to foresight and the scorer, while reducing the regression target from raw marker grids to a compact contact representation.

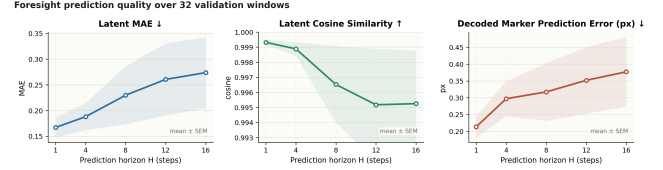


Fig. 1. Foresight prediction quality across horizons. Longer horizons have larger decoded marker error, but provide more task-relevant future contact context for action guidance.

B. Interpretation

The reconstruction table shows that the marker field is low-dimensional enough for simple baselines to reconstruct many static frames. This is why reconstruction alone is not the full argument for TacVAE. The relevant property for ForeTac is whether the latent is useful in the full differentiable chain: current tactile history $\rightarrow$ future tactile prediction $\rightarrow$ contact-quality score $\rightarrow$ action update. TacVAE is therefore evaluated together with foresight prediction and guidance diagnostics, not only as a standalone autoencoder.

VI. FORESIGHT PREDICTION QUALITY

The horizon study evaluates whether a longer predicted tactile future improves action selection despite the expected degradation in prediction error. Table V reports the current board-wiping numbers. The 16-step horizon is the main setting and matches the main online success rate used by the controller. Short horizons have lower reconstruction error but do not expose enough future contact consequence for stable guidance.

VII. CONTACT-QUALITY SCORER VALIDATION

A. Preference-Order Accuracy

The scorer is validated with positive/negative rollout pairs. For each pair, a key contact frame is selected from the offline rollout. The pair is counted as correct when the positive rollout receives a higher score than the negative rollout. This validates the ranking property required by the energy model: the absolute score does not need to be calibrated perfectly, but the positive contact segment should score above the failure segment.

TABLE IV

RECONSTRUCTION COMPARISON ACROSS TACTILE ENCODERS. EACH TASK REPORTS MAE / COSINE SIMILARITY. BEST AVAILABLE VALUES IN EACH TASK-METRIC COLUMN ARE BOLDED.

Encoder	Board		Vase		Card		Chip	
	MAE ↓	Cos. ↑	MAE ↓	Cos. ↑	MAE ↓	Cos. ↑	MAE ↓	Cos. ↑
PCA	0.040	0.995	0.035	0.995	0.036	0.980	0.026	0.965
PointNet-AE	0.048	0.956	0.027	0.955	0.044	0.992	0.032	0.990
Conv-AE	0.055	0.997	0.042	0.994	0.104	0.999	0.036	0.959
TacVAE (ours)	0.039	0.998	0.037	0.975	0.031	0.999	0.015	0.899

TABLE V

FORESIGHT PREDICTION QUALITY AND ROLLOUT SUCCESS ACROSS PREDICTION HORIZONS.

Horizon	Latent MAE ↓	Latent cosine ↑	Marker error ↓	Success ↑
1	0.167	0.9993	0.213 px	60%
4	0.188	0.9989	0.297 px	68%
8	0.230	0.9965	0.317 px	75%
12	0.261	0.9952	0.352 px	80%
16	0.274	0.9953	0.377 px	85%

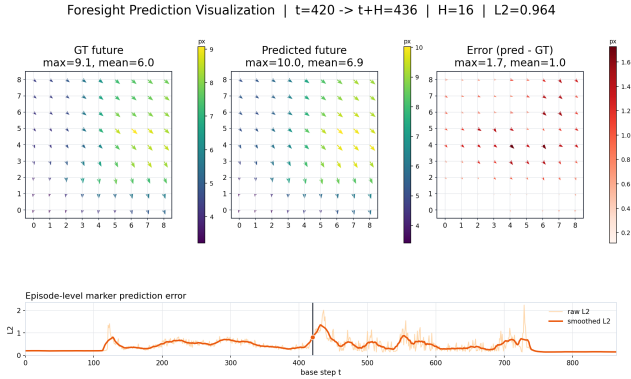


Fig. 2. Example long-horizon board prediction. The predicted marker field preserves the contact direction and spatial contact pattern needed by the scorer.

The key-frame and model-best selections are more meaningful than averaging all windows because most non-contact or low-contact windows are ambiguous and do not reflect the decisive contact event. The current smoke subset is small, so the table is evidence for scorer behavior rather than a formal final result. The next complete validation should use all rollout pairs collected under the same initial-condition protocol.

B. Gradient and Sampler Diagnostics

The scorer must also be usable as a differentiable guidance signal. The board stride-3 diagnostic verifies that the composed action-foresight-score chain produces finite gradients and bounded action updates.

VIII. FORCE-CURVE ANALYSIS

Force traces provide an interpretable view of the contact-quality objective. For wiping tasks, the desired behavior is not maximum force. The desired behavior is a stable force band that maintains contact without disturbing the object. For insertion, the desired behavior is bounded peak force plus few retries; a policy that keeps pushing after contact can look geometrically reasonable but is unsafe.

IX. MAIN COMPARISON

Table VIII is the current main rollout table. Each entry reports success rate and successful trials over 20 attempts. Safe success is the stricter subset that also satisfies the safety criteria defined in Section II.

The main comparison separates two improvements. First, tactile concatenation helps the base policy by adding current contact evidence, but it remains reactive. Second, ForeTac uses predicted tactile consequences before action execution, giving the scorer a way to penalize contact dropout, excessive force, and unstable contact earlier in the action chunk. The largest gains are on tasks where visual geometry is insufficient to infer contact quality: board wiping, chip grasping, and socket-style insertion.

X. ABLATION STUDY

Table IX isolates the role of tactile observations, foresight, and guided denoising. DP Vision uses only visual/proprioceptive observations. TacConcat adds current tactile latent features to the policy. Only Foresight uses future tactile prediction without the full denoising guidance route. DP Guide is the complete predict-score-guide system.

The ablation suggests that current tactile input is useful but insufficient: TacConcat improves over vision-only DP, while Only Foresight improves safety on several tasks by exposing future contact. The full DP Guide variant gives the largest improvement because it uses foresight not only for representation, but also as a differentiable contact-quality objective during late denoising.

XI. INSERTION BENCHMARK

The socket insertion benchmark uses 20 trials per method. Besides success, the task records retry count, maximum

TABLE VI
OFFLINE PREFERENCE-ORDER VALIDATION ON THE CURRENT BOARD SMOKE ROLLOUT SUBSET.

Selection rule	Score type	Correct / pairs	Accuracy	Mean gap	Median gap
Key marker frame	expert margin	2 / 3	0.6667	0.033707	0.024794
Key marker frame	quality 0–100	2 / 3	0.6667	0.000956	0.000685
Key force frame	expert margin	2 / 3	0.6667	-0.012368	0.016713
Model-best window	expert margin	2 / 3	0.6667	0.060702	0.092206
Model-best window	quality 0–100	2 / 3	0.6667	0.001863	0.002820
All windows	expert margin	1090 / 2436	0.4475	-0.003016	-0.006750
All windows	quality 0–100	1090 / 2436	0.4475	-0.000048	-0.000190

Blackboard Guidance Diagnostics

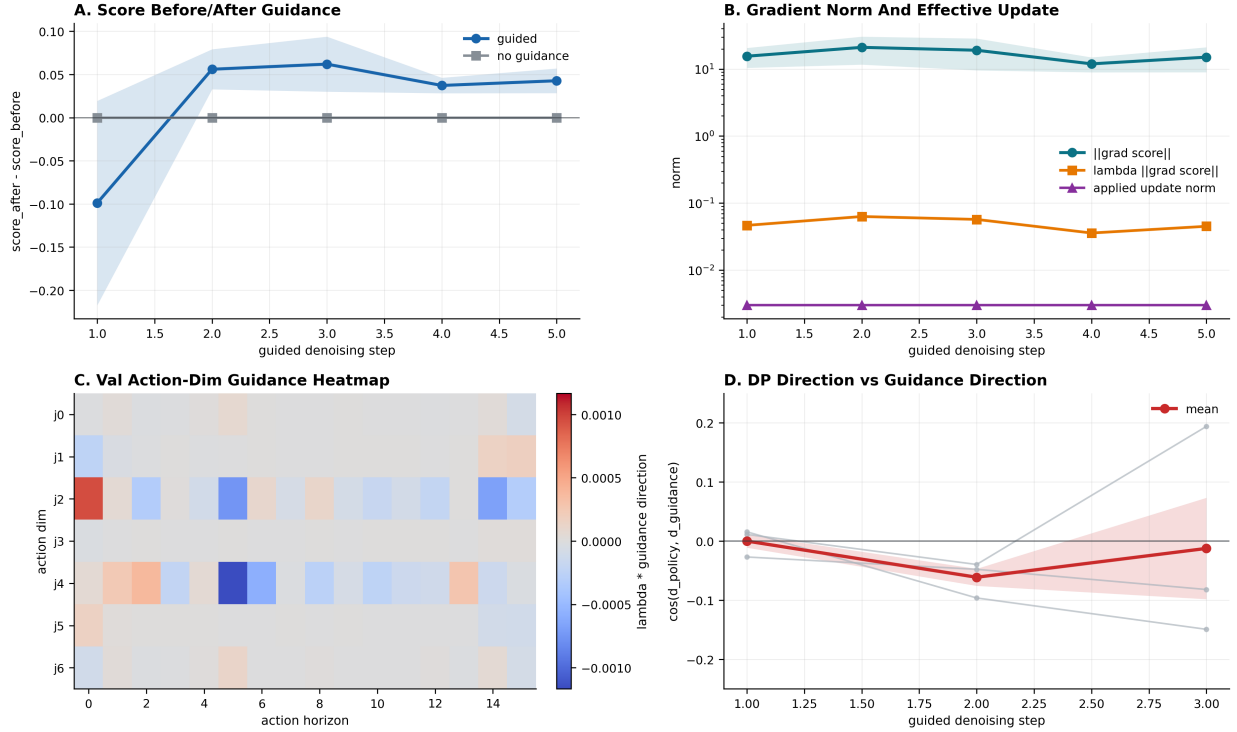


Fig. 3. Board guidance visualizations. The panels summarize score changes, gradient magnitude, action-dimension localization, and policy-guidance alignment. The update is finite and localized, consistent with a small contact correction rather than a full trajectory rewrite.

observed contact force, mean peak force, and recovery time from peak force to below 2 N.

Raw DP has no explicit recovery behavior, so “retry” is reported as zero in failed no-lift trials rather than as a sign of stable insertion. DP+Tac can recover, but the recovery appears as repeated high-force contact cycles. Foresight-guided variants reduce both the number of retries and the magnitude of force peaks.

XII. BACKBONE ADAPTATION

ForeTac is designed to wrap any chunk-based action generator. For DP, the guidance gradient is applied directly to the denoising action sample. For RDP, the same predicted-contact score is attached to the reactive diffusion transformer

action chunk. For $\pi_{0.5}$, the current adaptation keeps the VLA backbone in a flow-matching action-generation role and appends tactile latent tokens to the policy prefix; the foresight branch remains the action-conditioned contact consequence model. In all cases, the contact-quality scorer is outside the base policy and can be evaluated independently.

XIII. OPEN ITEMS FOR FINAL APPENDIX

The following items should be filled before camera-ready submission:

- Replace smoke preference-order accuracy with the full paired rollout validation once all positive/negative pairs are collected.

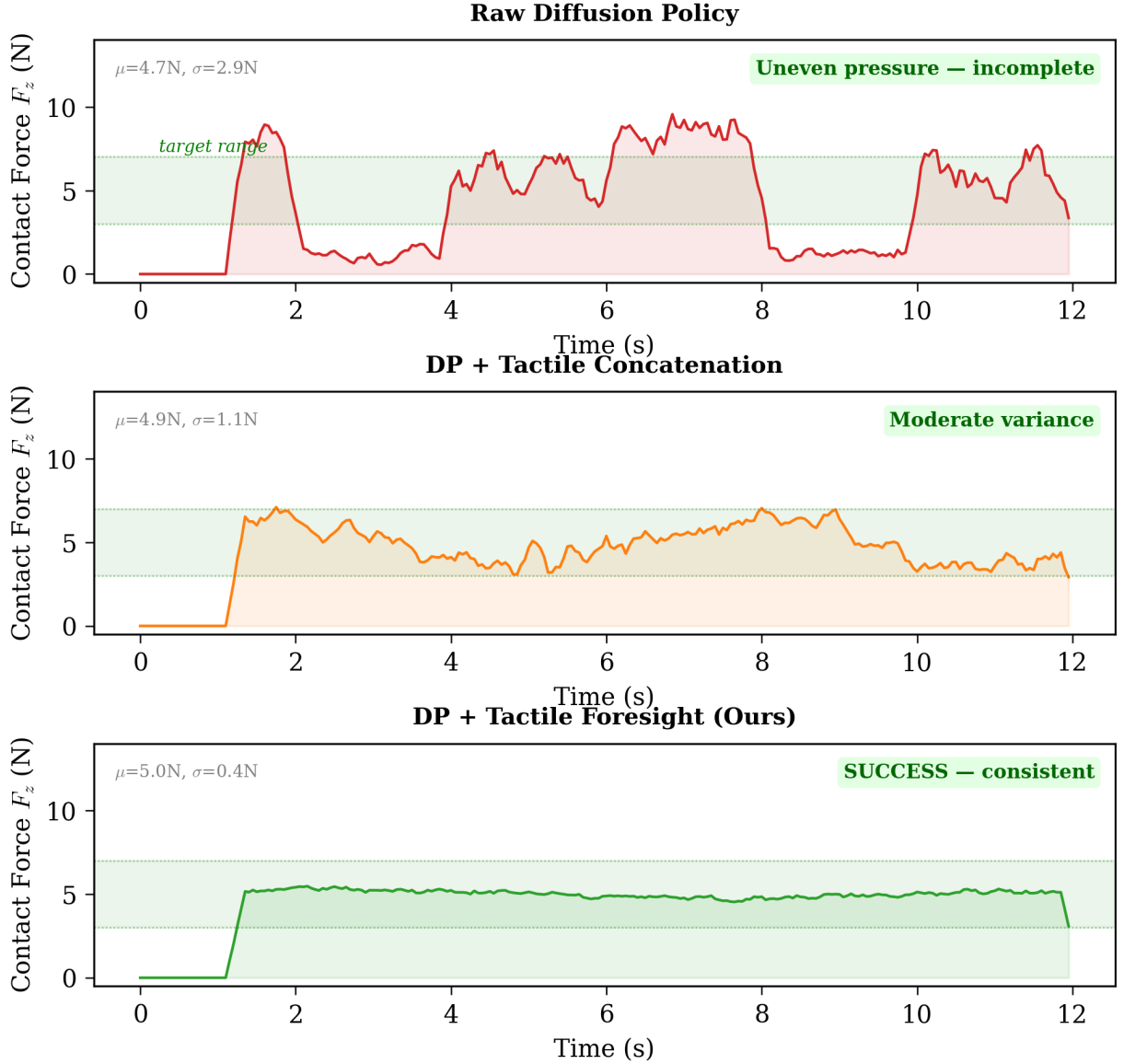


Fig. 4. Board wiping force traces. Raw DP produces unstable pressure and contact dropouts. Tactile concatenation improves the mean force but still exhibits moderate variation. ForeTac keeps the force in the desired band with substantially lower variance.

TABLE VII
BOARD STRIDE-3 GUIDANCE DIAGNOSTIC.

Metric	Value
Validation windows	3144
Scorer accuracy / macro-F1	1.000 / 1.000
Expert-margin AUROC	1.000
Rows with finite gradients	128 / 128
Mean raw gradient norm	2.4475
Mean scaled gradient norm	0.00734
Mean guided-step score delta	0.0198
Mean applied update norm	0.0030

of the main comparison.

- Add qualitative panels for all five tasks using the final visual assets from the project webpage.
- Re-run the horizon table after the final action-conditioned foresight checkpoint is selected, and keep the action-conditioned config archived with the appendix source.

- Fill the chip grasping data inventory and add the missing chip force curve if it is used in the final paper.
- Add final $\pi_{0.5}$ and ForeAR rollout rows if they are part

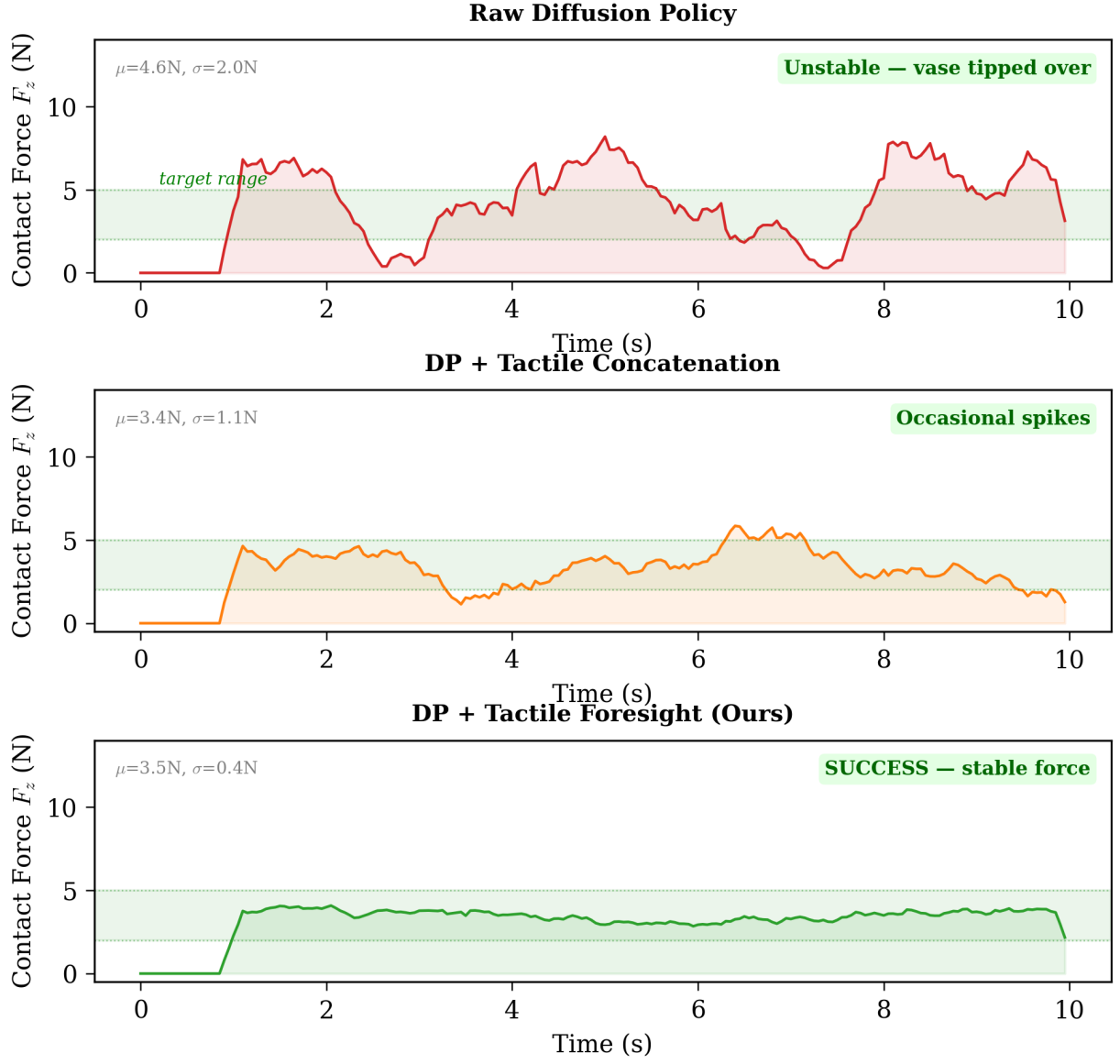


Fig. 5. Vase wiping force traces. The curved fragile surface makes force stability more important than on a flat board. The guided policy maintains a lower-variance force profile and avoids the tip-over mode observed for raw DP.

TABLE VIII
MAIN COMPARISON ACROSS CONTACT-RICH MANIPULATION TASKS.

Method	Board Succ.	Board Safe	Vase Succ.	Vase Safe	Vase Up-Down Succ.	Vase Up-Down Safe	Card Succ.	Card Safe	Chip Succ.	Chip Safe
DP	50% (10/20)	25% (5/20)	55% (11/20)	35% (7/20)	5% (1/20)	0% (0/20)	10% (2/20)	5% (1/20)	30% (6/20)	20% (4/20)
DP+tactile	60% (12/20)	50% (10/20)	60% (12/20)	50% (10/20)	40% (8/20)	30% (6/20)	35% (7/20)	30% (6/20)	40% (8/20)	35% (7/20)
$\pi_{0.5}$	55% (11/20)	45% (9/20)	60% (12/20)	45% (9/20)	30% (6/20)	20% (4/20)	25% (5/20)	20% (4/20)	45% (9/20)	35% (7/20)
RDP	65% (13/20)	55% (11/20)	65% (13/20)	55% (11/20)	45% (9/20)	35% (7/20)	40% (8/20)	35% (7/20)	50% (10/20)	45% (9/20)
Ours	85% (17/20)	80% (16/20)	70% (14/20)	65% (13/20)	55% (11/20)	50% (10/20)	55% (11/20)	50% (10/20)	75% (15/20)	70% (14/20)

TABLE IX
ABLATION STUDY. EACH ENTRY REPORTS SUCCESS OVER 20 ATTEMPTS AND SAFE SUCCESS OVER THE SAME 20 ATTEMPTS.

Variant	Board Succ.	Board Safe	Vase Succ.	Vase Safe	Vase Up-Down Succ.	Vase Up-Down Safe	Card Succ.	Card Safe	Chip Succ.	Chip Safe
DP Vision	50% (10/20)	35% (7/20)	55% (11/20)	35% (7/20)	10% (2/20)	5% (1/20)	10% (2/20)	5% (1/20)	30% (6/20)	20% (4/20)
TacConcat	60% (12/20)	50% (10/20)	60% (12/20)	50% (10/20)	40% (8/20)	30% (6/20)	35% (7/20)	30% (6/20)	40% (8/20)	35% (7/20)
Only Foresight	60% (12/20)	55% (11/20)	65% (13/20)	55% (11/20)	45% (9/20)	35% (7/20)	40% (8/20)	35% (7/20)	55% (11/20)	50% (10/20)
DP Guide	85% (17/20)	80% (16/20)	70% (14/20)	65% (13/20)	55% (11/20)	50% (10/20)	55% (11/20)	50% (10/20)	75% (15/20)	70% (14/20)

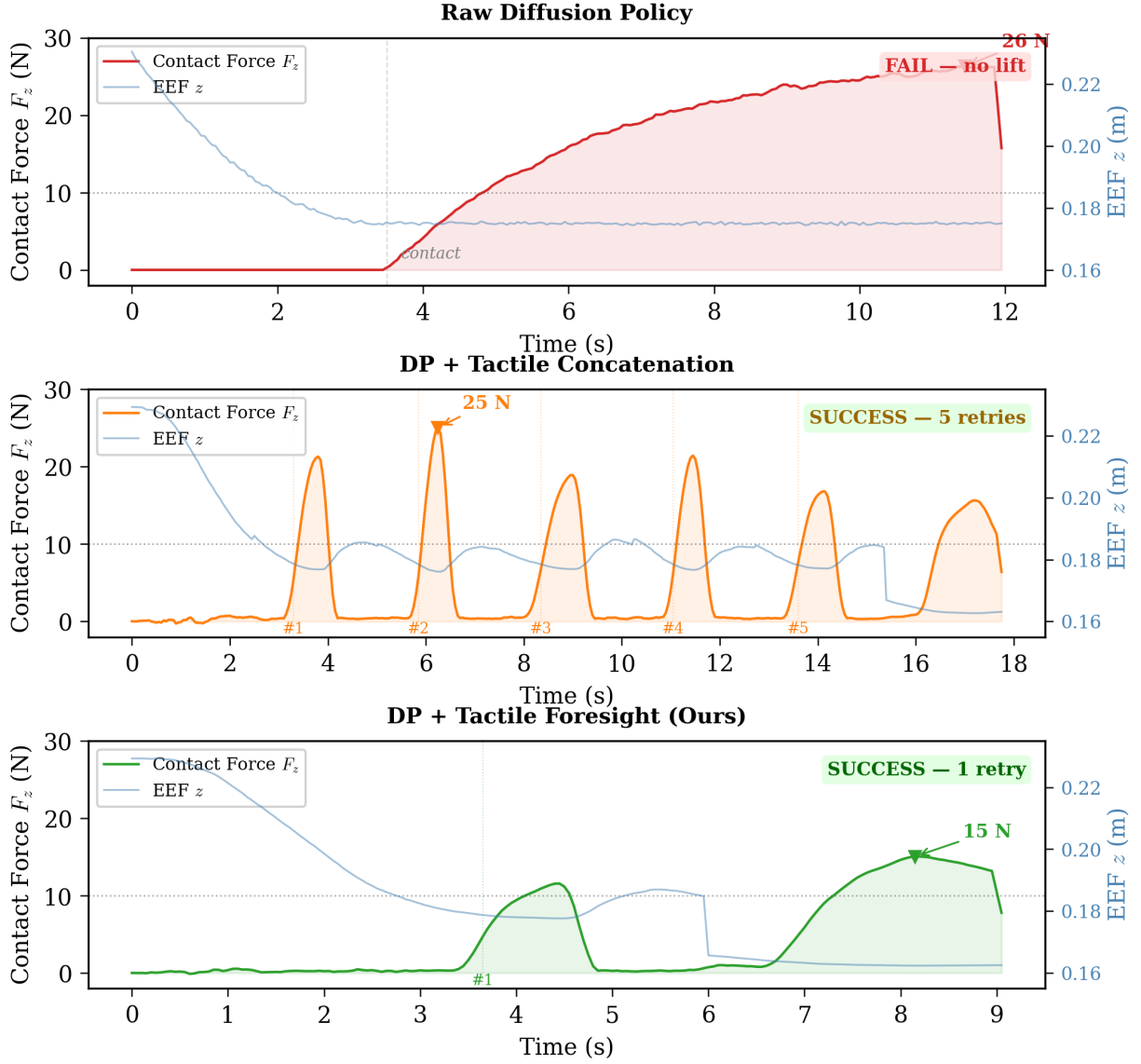


Fig. 6. Socket insertion force traces. Raw DP sustains high force and fails to lift after contact; tactile concatenation succeeds with repeated bounce-lift-reinsert cycles; ForeTac reduces peak force and recovery time.

TABLE X
PEG-IN-HOLE / SOCKET INSERTION BENCHMARK.

Method	Succ. $\uparrow$	Retry $\downarrow$	Max F (N) $\downarrow$	Peak F (N) $\downarrow$	Recover (s) $\downarrow$
Raw DP	35% (7/20)	0	28.1	24.3	—
DP+Tac	60% (12/20)	4.2	26.4	21.5	1.85
DP+Foresight	85% (17/20)	1.1	15.2	12.8	0.42
RDT+Foresight	90% (18/20)	0.9	14.1	11.5	0.35

TABLE XI
BACKBONE ADAPTATION SUMMARY.

Backbone	Adaptation	Additional policy training
DP	late denoising action update	task DP checkpoint
RDP	same score on reactive action chunk	task RDP checkpoint
$\pi_{0.5}$	tactile latent prefix + flow-matching action chunk	$\pi_{0.5}$ fine-tuning / adapter
ForeAR	autoregressive foresight baseline	TBD